

Online Supplement to “On the Network Coding Capacity of Wireless Multicast Networks under One-Hop Interference”

Xiaohong Cai and Raymond W. Yeung

1 Finite Matching-Based Verification

For each reduced butterfly-like configuration, we construct the associated directed hypergraph $H = (\mathcal{N}, \mathcal{E}_H)$. A feasible simultaneous transmission set is a matching of H , i.e., a set of hyperedges no two of which are incident to a common node. Let $\mathcal{F}_H$ denote the set of all matchings of H , including the empty matching, and let ω_σ be the fraction of the resource assigned to $\sigma \in \mathcal{F}_H$.

For each destination $t \in \{X, Y\}$, let $\mathcal{C}_H^{st}$ denote the set of all s - t cutsets in H , as defined in the main paper. Thus, for an s - t cut with source side $\mathcal{S} \subseteq \mathcal{N}$, where $s \in \mathcal{S}$ and $t \in \mathcal{N} \setminus \mathcal{S}$, the associated cutset contains every hyperedge $(u, V) \in \mathcal{E}_H$ such that $u \in \mathcal{S}$ and $V \cap (\mathcal{N} \setminus \mathcal{S}) \neq \emptyset$.

Following the notation for butterfly-like networks in the main paper, the subpaths P^{AI} , P^{BI} , P^{DX} , P^{DY} , and P^{ID} are denoted by c , d , g , h , and i , respectively. We define $\mathcal{E}_{\text{inner}}$ as the set of all directed edges on c , d , g , and h , together with those on i when $I \neq D$.

For each reduced configuration, we maximize the multicast rate R over the matching weights $\{\omega_\sigma\}_{\sigma \in \mathcal{F}_H}$. Let $C_{(u,V)}$ denote the capacity of hyper-

edge (u, V) induced by these weights. We solve the following linear program:

$$\begin{aligned}
& \text{maximize} && R \\
& \text{subject to} && \sum_{\sigma \in \mathcal{F}_H} \omega_\sigma = 1, \\
& && C_{(u,V)} = \sum_{\sigma \in \mathcal{F}_H: (u,V) \in \sigma} \omega_\sigma, \quad (u, V) \in \mathcal{E}_H, \\
& && R \leq \sum_{(u,V) \in K} C_{(u,V)}, \quad K \in \mathcal{C}_H^{st}, \quad t \in \{X, Y\}, \\
& && \sum_{\substack{\sigma \in \mathcal{F}_H: \exists (u,V) \in \sigma, \\ v \in V}} \omega_\sigma = \frac{1}{3}, \quad (u, v) \in \mathcal{E}_{\text{inner}}, \\
& && \omega_\sigma \geq 0, \quad \sigma \in \mathcal{F}_H, \\
& && 0 \leq R \leq 1.
\end{aligned}$$

The equality constraint indexed by $(u, v) \in \mathcal{E}_{\text{inner}}$ fixes the induced capacity of every directed edge on an inner subpath to $1/3$.

The exhaustive computation verifies every reduced configuration for both $I = D$ and $I \neq D$. In each case, the configurations with LP value at least $5/6$ are exactly those satisfying the corresponding structural conditions in the main paper. Together with the $5/6$ upper bound for butterfly-like networks, this shows that every reduced configuration satisfying these conditions has capacity exactly $5/6$. The implementation and complete machine-readable output are provided with this supplement.

2 Proof of Theorem 8

Let $\mathcal{T} = \{X, Y\}$. Without loss of generality, assume that G_0 contains two disjoint routing paths P_X and $\bar{P}_X$ from s to X , and three disjoint routing paths Q_1 , Q_2 , and Q_3 from s to Y . We use the subpath notation P^{uv} defined in Section III in the main paper. For ease of notation, when $u = v$, P^{uv} is omitted in path constructions.

Since G_0 is a directed acyclic graph, its nodes admit a topological ordering. Fix one such ordering and denote it by “ $<$ ”. For any two distinct nodes $u, v \in N(G_0)$, if there is a directed path from u to v , then $u < v$. Throughout the proof, this ordering is used both to compare nodes and to identify the last node in a node set.

The proof below constructs four routing paths: two disjoint paths connecting s to X , and two disjoint paths connecting s to Y . For each con-

struction, let G_r be the subgraph whose node set is the union of the node sets of the four selected paths and whose edge set is the union of their edge sets. By Definition 4, G_r is a routing graph. Since all selected paths lie in G_0 , G_r is a routing subgraph of G_0 . It remains to show that this routing subgraph contains no intersection node. Since the two paths to the same destination are disjoint by construction, we only need to check those pairs consisting of one path to X and one path to Y . By Propositions 1 and 2, it is enough to verify that each such pair is either disjoint or a multicast.

For each $i \in \{1, 2, 3\}$, let M_i be the last node in

$$N(Q_i) \cap (N(P_X) \cup N(\bar{P}_X)). \quad (1)$$

If $M_i \neq s$, then M_i lies on exactly one of P_X and $\bar{P}_X$, since P_X and $\bar{P}_X$ are disjoint except at s and X , and a routing path to Y cannot pass through the destination X .

We distinguish three cases according to how many of M_1 , M_2 , and M_3 are equal to s and whether the remaining nodes lie on the same path in $\{P_X, \bar{P}_X\}$.

Case 1. At least two of M_1 , M_2 , and M_3 are equal to s . Without loss of generality, let $M_1 = M_2 = s$. Then neither Q_1 nor Q_2 has a node in $N(P_X) \cup N(\bar{P}_X)$ other than s . Thus

$$P_X, \bar{P}_X, Q_1, Q_2$$

already form the required four paths. Every pair connecting to different destinations is disjoint, so the subgraph formed by these four paths contains no intersection node.

Case 2. Case 1 does not occur, and there exist two non-source nodes among M_1 , M_2 , and M_3 that lie on different paths in $\{P_X, \bar{P}_X\}$. That is, there exist $i \neq j$ such that $M_i, M_j \neq s$, with M_i and M_j lying on different paths in $\{P_X, \bar{P}_X\}$. Without loss of generality, assume that $M_i \in N(P_X)$ and $M_j \in N(\bar{P}_X)$. Define

$$Q'_i = (P_X^{sM_i}, Q_i^{M_iY}), \quad Q'_j = (\bar{P}_X^{sM_j}, Q_j^{M_jY}).$$

Disjointness of Q'_i and Q'_j : The subpaths $P_X^{sM_i}$ and $\bar{P}_X^{sM_j}$ are disjoint because P_X and $\bar{P}_X$ are disjoint. Likewise, $Q_i^{M_iY}$ and $Q_j^{M_jY}$ are disjoint because Q_i and Q_j are disjoint. It remains to consider the two cross-pairs. By the definition of M_i , $Q_i^{M_iY}$ shares no node with $\bar{P}_X^{sM_j}$; similarly, by the

definition of M_j , $Q_j^{M_j Y}$ shares no node with $P_X^{sM_i}$. Thus, Q'_i and Q'_j are disjoint.

Multicast pairs formed by P_X and Q'_i , and by $\bar{P}_X$ and Q'_j : The paths P_X and Q'_i share the initial subpath $P_X^{sM_i}$, and $Q_i^{M_i Y}$ shares no further node with P_X by the definition of M_i . Thus, P_X and Q'_i form a multicast. Similarly, $\bar{P}_X$ and Q'_j form a multicast with common prefix $\bar{P}_X^{sM_j}$.

Disjointness of P_X from Q'_j and of $\bar{P}_X$ from Q'_i : The subpath $\bar{P}_X^{sM_j}$ is disjoint from P_X , and $Q_j^{M_j Y}$ shares no node with P_X by the definition of M_j . Similarly, the subpath $P_X^{sM_i}$ is disjoint from $\bar{P}_X$, and $Q_i^{M_i Y}$ shares no node with $\bar{P}_X$ by the definition of M_i .

Consequently, P_X , $\bar{P}_X$, Q'_i , and Q'_j form a routing subgraph of G_0 , and every pair of paths connecting to different destinations is either disjoint or a multicast. By Propositions 1 and 2, this routing subgraph contains no intersection node.

Case 3. Neither Case 1 nor Case 2 occurs. Since Case 1 does not occur, at most one of M_1 , M_2 , and M_3 is s . Since Case 2 does not occur, all M_i that are non-source nodes lie on the same path in $\{P_X, \bar{P}_X\}$. Moreover, these non-source nodes are distinct because Q_1 , Q_2 , and Q_3 are disjoint (cf. (1)). Without loss of generality, assume that this same path is P_X , and relabel Q_1 , Q_2 , and Q_3 so that

$$M_1 < M_2 < M_3 \quad \text{on } P_X,$$

where $M_1 = s$ is allowed.

For each i , let L_i be the last node in

$$N(Q_i) \cap N(\bar{P}_X). \quad (2)$$

If Q_i and $\bar{P}_X$ share no non-source node, then $L_i = s$.

Subcase 3a. There exist $i < j$ such that $L_j \leq L_i$. Define

$$P'_X = (Q_j^{sM_j}, P_X^{M_j X}), \quad \bar{P}'_X = (Q_i^{sL_i}, \bar{P}_X^{L_i X}),$$

Disjointness of P'_X and $\bar{P}'_X$: The subpaths $Q_j^{sM_j}$ and $Q_i^{sL_i}$ are disjoint because Q_j and Q_i are disjoint. Likewise, $P_X^{M_j X}$ and $\bar{P}_X^{L_i X}$ are disjoint because P_X and $\bar{P}_X$ are disjoint. It remains to consider the two cross-pairs. Since $M_i < M_j$ and M_i is the last node in $N(Q_i) \cap N(P_X)$, we have $N(Q_i) \cap N(P_X^{M_j X}) = \emptyset$. Therefore, $Q_i^{sL_i}$ shares no node with $P_X^{M_j X}$. Since

Q_i and Q_j are disjoint, $L_i = L_j$ can occur only when $L_i = L_j = s$. Together with $L_j \leq L_i$ and the fact that L_j is the last node in $N(Q_j) \cap N(\bar{P}_X)$, this gives

$$(N(Q_j) \setminus \{s\}) \cap N(\bar{P}_X^{L_i X}) = \emptyset.$$

Therefore, $Q_j^{sM_j}$ shares no node with $\bar{P}_X^{L_i X}$ other than possibly s . Thus, P'_X and $\bar{P}'_X$ are disjoint.

Disjointness of $\bar{P}'_X$ from Q_j and of P'_X from Q_i : Q_j is disjoint from $Q_i^{sL_i}$, and $(N(Q_j) \setminus \{s\}) \cap N(\bar{P}_X^{L_i X}) = \emptyset$, as shown above. Moreover, Q_i is disjoint from $Q_j^{sM_j}$, and $N(Q_i) \cap N(P_X^{M_j X}) = \emptyset$, as shown above.

Multicast pair formed by P'_X and Q_j : The paths P'_X and Q_j share the initial subpath $Q_j^{sM_j}$, and $Q_j^{M_j Y}$ shares no further node with P_X by the definition of M_j (cf. (1)). Thus, P'_X and Q_j form a multicast.

Analysis of $\bar{P}'_X$ and Q_i : If $L_i \neq s$, then $\bar{P}'_X$ and Q_i share the initial subpath $Q_i^{sL_i}$, and the remainder of Q_i shares no further node with $\bar{P}_X$ by the definition of L_i (cf. (2)); hence, they form a multicast. If $L_i = s$, then $\bar{P}'_X$ and Q_i are disjoint.

Since Q_i and Q_j are disjoint, P'_X , $\bar{P}'_X$, Q_i , and Q_j form a routing subgraph of G_0 , and every pair of paths connecting to different destinations is either disjoint or a multicast. By Propositions 1 and 2, this routing subgraph contains no intersection node.

Subcase 3b. No pair $i < j$ satisfies $L_j \leq L_i$. This implies

$$L_1 < L_2 < L_3 \quad \text{on } \bar{P}_X.$$

Consider $Q_3^{sM_3}$, where $M_3 \in N(P_X)$, and let $A_0 = M_3$. By tracing Q_3 backwards from A_0 , we will construct nodes

$$A_0, A_1, A_2, \dots, A_l \in N(P_X) \cap N(Q_3^{sM_3})$$

and

$$B_1, B_2, \dots, B_l \in N(\bar{P}_X) \cap N(Q_3^{sM_3}),$$

where $l \geq 1$, such that

$$M_3 = A_0 > B_1 > A_1 > B_2 > \dots > A_{l-1} > B_l \geq A_l = s.$$

Specifically, given $A_{k-1} \neq s$, let B_k be the last node in $N(Q_3^{sA_{k-1}}) \cap N(\bar{P}_X)$.¹ Then let A_k be the last node in $N(Q_3^{sB_k}) \cap N(P_X)$.² By construction, for

¹If $Q_3^{sA_{k-1}}$ and $\bar{P}_X$ share no non-source node, then $B_k = s$.

²If $Q_3^{sB_k}$ and P_X share no non-source node, then $A_k = s$.

each k with $A_{k-1} \neq s$, we have $A_{k-1} > B_k \geq A_k$, where equality can occur only when $B_k = A_k = s$. Consequently, $A_0 > A_1 > A_2 > \dots > A_l = s$ on P_X . For this sequence, since

1. $A_0 = M_3 > M_2$;
2. the sequence eventually reaches s ;
3. no A_k can equal M_2 since Q_2 and Q_3 are disjoint (note that $A_k \in N(Q_3)$ and $M_2 \in N(Q_2)$),

there exists an $r \geq 1$ such that

$$A_r < M_2 < A_{r-1} \quad \text{on } P_X.$$

Since $L_1 < L_2$, we have $L_2 \neq s$. Moreover, $B_r \neq L_2$ because equality would make Q_2 and Q_3 share the non-source node L_2 (note that $B_r \in N(Q_3)$ and $L_2 \in N(Q_2)$). Therefore, either $B_r > L_2$ or $B_r < L_2$ on $\bar{P}_X$.

Subcase 3b(i). $B_r > L_2$. Define

$$P'_X = (Q_2^{sM_2}, P_X^{M_2X}), \quad \bar{P}'_X = (Q_3^{sB_r}, \bar{P}_X^{B_rX}),$$

Disjointness of P'_X and $\bar{P}'_X$: The subpaths $Q_2^{sM_2}$ and $Q_3^{sB_r}$ are disjoint because Q_2 and Q_3 are disjoint. Likewise, $P_X^{M_2X}$ and $\bar{P}_X^{B_rX}$ are disjoint because P_X and $\bar{P}_X$ are disjoint. It remains to consider the two cross-pairs. Since $L_2 < B_r$ on $\bar{P}_X$ and L_2 is the last node in $N(Q_2) \cap N(\bar{P}_X)$, we have $N(Q_2) \cap N(\bar{P}_X^{B_rX}) = \emptyset$. Therefore, $Q_2^{sM_2}$ shares no node with $\bar{P}_X^{B_rX}$. Since $A_r < M_2$ and A_r is the last node in $N(Q_3^{sB_r}) \cap N(P_X)$, we have $N(Q_3^{sB_r}) \cap N(P_X^{M_2X}) = \emptyset$. Therefore, $Q_3^{sB_r}$ shares no node with $P_X^{M_2X}$. Thus, P'_X and $\bar{P}'_X$ are disjoint.

Disjointness of $\bar{P}'_X$ from Q_2 : Q_2 is disjoint from $Q_3^{sB_r}$. Moreover, $N(Q_2) \cap N(\bar{P}_X^{B_rX}) = \emptyset$ as shown above.

Disjointness of P'_X from Q_1 : Q_1 is disjoint from $Q_2^{sM_2}$. Moreover, $N(Q_1) \cap N(P_X^{M_2X}) = \emptyset$, since $M_1 < M_2$ and M_1 is the last node in $N(Q_1) \cap N(P_X)$.

Disjointness of $\bar{P}'_X$ from Q_1 : Q_1 is disjoint from $Q_3^{sB_r}$. Moreover, $N(Q_1) \cap N(\bar{P}_X^{B_rX}) = \emptyset$, since $L_1 < L_2 < B_r$ and L_1 is the last node in $N(Q_1) \cap N(\bar{P}_X)$.

Multicast pair formed by P'_X and Q_2 : The paths P'_X and Q_2 share the initial subpath $Q_2^{sM_2}$, and $Q_2^{M_2Y}$ shares no further node with P_X by the definition of M_2 (cf. (1)). Thus, P'_X and Q_2 form a multicast.

Since Q_2 and Q_1 are disjoint, P'_X , $\bar{P}'_X$, Q_2 , and Q_1 form a routing subgraph of G_0 , and every pair of paths connecting to different destinations is either disjoint or a multicast. By Propositions 1 and 2, this routing subgraph contains no intersection node.

Subcase 3b(ii). $B_r < L_2$. Define

$$P'_X = (Q_3^{sA_{r-1}}, P_X^{A_{r-1}X}), \quad \bar{P}'_X = (Q_2^{sL_2}, \bar{P}_X^{L_2X}),$$

Disjointness of P'_X and $\bar{P}'_X$: The subpaths $Q_3^{sA_{r-1}}$ and $Q_2^{sL_2}$ are disjoint because Q_3 and Q_2 are disjoint. Likewise, $P_X^{A_{r-1}X}$ and $\bar{P}_X^{L_2X}$ are disjoint because P_X and $\bar{P}_X$ are disjoint. It remains to consider the two cross-pairs. Since $B_r < L_2$ on $\bar{P}_X$ and B_r is the last node in $N(Q_3^{sA_{r-1}}) \cap N(\bar{P}_X)$, we have $N(Q_3^{sA_{r-1}}) \cap N(\bar{P}_X^{L_2X}) = \emptyset$. Therefore, $Q_3^{sA_{r-1}}$ shares no node with $\bar{P}_X^{L_2X}$. Since $M_2 < A_{r-1}$ and M_2 is the last node in $N(Q_2) \cap N(P_X)$, we have $N(Q_2) \cap N(P_X^{A_{r-1}X}) = \emptyset$. Therefore, $Q_2^{sL_2}$ shares no node with $P_X^{A_{r-1}X}$. Thus, P'_X and $\bar{P}'_X$ are disjoint.

Disjointness of P'_X from Q_2 : Q_2 is disjoint from $Q_3^{sA_{r-1}}$, and $N(Q_2) \cap N(P_X^{A_{r-1}X}) = \emptyset$, as shown above.

Disjointness of P'_X from Q_1 : Q_1 is disjoint from $Q_3^{sA_{r-1}}$, and $N(Q_1) \cap N(P_X^{A_{r-1}X}) = \emptyset$, since $M_1 < M_2 < A_{r-1}$ and M_1 is the last node in $N(Q_1) \cap N(P_X)$.

Disjointness of $\bar{P}'_X$ from Q_1 : Q_1 is disjoint from $Q_2^{sL_2}$. Moreover, $N(Q_1) \cap N(\bar{P}_X^{L_2X}) = \emptyset$, since $L_1 < L_2$ and L_1 is the last node in $N(Q_1) \cap N(\bar{P}_X)$.

Multicast pair formed by $\bar{P}'_X$ and Q_2 : The paths $\bar{P}'_X$ and Q_2 share the initial subpath $Q_2^{sL_2}$, and the remainder of Q_2 shares no further node with $\bar{P}_X$ by the definition of L_2 (cf. (2)). Thus, $\bar{P}'_X$ and Q_2 form a multicast.

Since Q_2 and Q_1 are disjoint, P'_X , $\bar{P}'_X$, Q_2 , and Q_1 form a routing subgraph of G_0 , and every pair of paths connecting to different destinations is either disjoint or a multicast. By Propositions 1 and 2, this routing subgraph contains no intersection node.

In all cases, we obtain two disjoint routing paths in G_0 to each destination, and the subgraph formed by these paths contains no intersection node. This completes the proof.